Physical security of fixed-point FN-DSA (Falcon)

Graduate Internship (Master 2), Summer 2027

Location	PQShield SAS, Paris, France 8 Rue des Pirogues de Bercy
Contact	Sebastien Riou, Thomas Prest
Starting date, duration	Q1-Q2 2027 (flexible), 6 months
Gross salary	1350 € / month

Internship

Background

Falcon was selected by NIST for standardization as FN-DSA (FIPS 206). It offers the most compact signatures among standardized post-quantum schemes. Its main drawback has long been its reliance on floating-point arithmetic, which is hard to implement in constant time and even harder to protect against side-channel attacks.

Recent work, including a fixed-point implementation of Falcon co-developed by PQShield researchers (CRYPTO 2026), has led NIST to announce plans to specify **fixed-point arithmetic for both key generation and signing in FIPS 206**. Fixed-point Falcon is about to become a standard, yet its physical security remains largely unexplored: it removes some weaknesses of floating-point arithmetic, but it comes with its own implementation trade-offs, and protecting it efficiently is an open problem.

This internship aims to understand how fixed-point FN-DSA behaves against physical attacks, and to design efficient ways to protect it.

Objectives

The main objectives of this internship project are to:

1. Understand Falcon / FN-DSA, its fixed-point signing procedure as planned for FIPS 206, and the state of the art in physical attacks and countermeasures on Falcon.
2. Deploy a fixed-point implementation on an ARM Cortex-M and RISC-V MCU. Assess its physical leakage, using evaluation methods and attacks from the literature.

3. Design and implement countermeasures (e.g., masking, shuffling) adapted to fixed-point arithmetic, and quantify their cost.
4. Explore new attack or protection techniques. The results are expected to lead to publications.

About PQShield

PQShield is a cybersecurity scaleup that specialises in post-quantum cryptography, protecting information from today's attacks while preparing organisations for the threat landscape of tomorrow. It demonstrates quantum-safe cryptography on chips, in applications and in the cloud. We are headquartered in Oxford, with additional teams in the Netherlands, Germany and France.

PQShield SAS, based in Paris (France), concentrates the research activities of PQShield. Our mission is to come up with innovative algorithmic and/or protocol-level solutions to real-world cryptographic problems. Besides post-quantum cryptographic primitives, our research interests include implementation security and advanced cryptosystems and protocols such as secure messaging, threshold schemes, and multiparty computation.

Requirements and how to apply

Ideal candidates should possess as many of these qualities as possible:

- Proficiency in C programming, and interest in embedded systems.
- Familiarity with lattice-based cryptography.
- Experience in side-channel analysis or with Python for data analysis is a plus.
- Strong autonomy.
- Good communication skills.

Prior knowledge of Falcon is not required.

Supervision: The intern will be supervised by Thomas Prest (Lead Cryptography Researcher — co-designer of Falcon) and Sebastien Riou (Fellow Product Security Architect). The intern will have access to PQShield's side-channel evaluation platform, on site in Paris or remotely.

References

1. Falcon: Fast-Fourier Lattice-based Compact Signatures over NTRU
2. Toward a Secure Fixed-Point Implementation of the Falcon Signature Scheme
3. The Hidden Parallelepiped Is Back Again: Power Analysis Attacks on Falcon